



SEGURANÇA EM REDES DE COMPUTADORES

(MIETI 4º ANO/S1 - 6707N5)

REVISITING TCP/IP / REVISÕES SOBRE O TCP/IP

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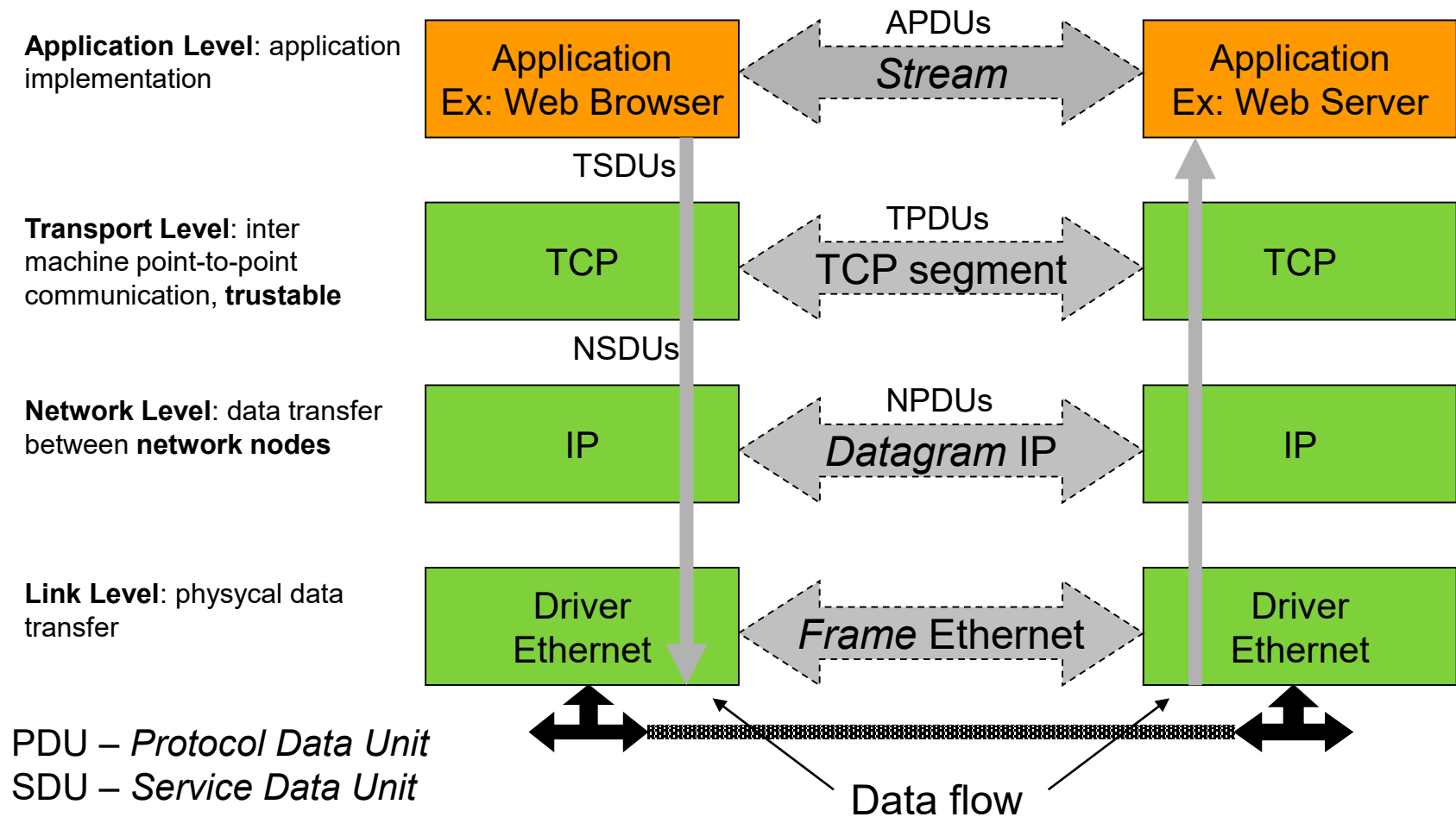


Summary

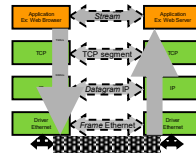
- Protocol stack
- Encapsulation
- Ports and addresses
- Supporting protocols
- *Routing*
- 3 phase *Handshaking*: illustration with Wireshark
- **Anomaly signs**



TCP/IP –Internet Communication model



TCP/IP –Internet Communication model



- Encapsulation

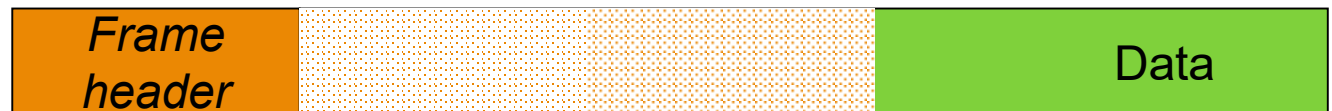
- TCP Segment (TPDU)



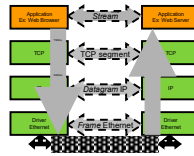
- Datagram IP (NPDU)



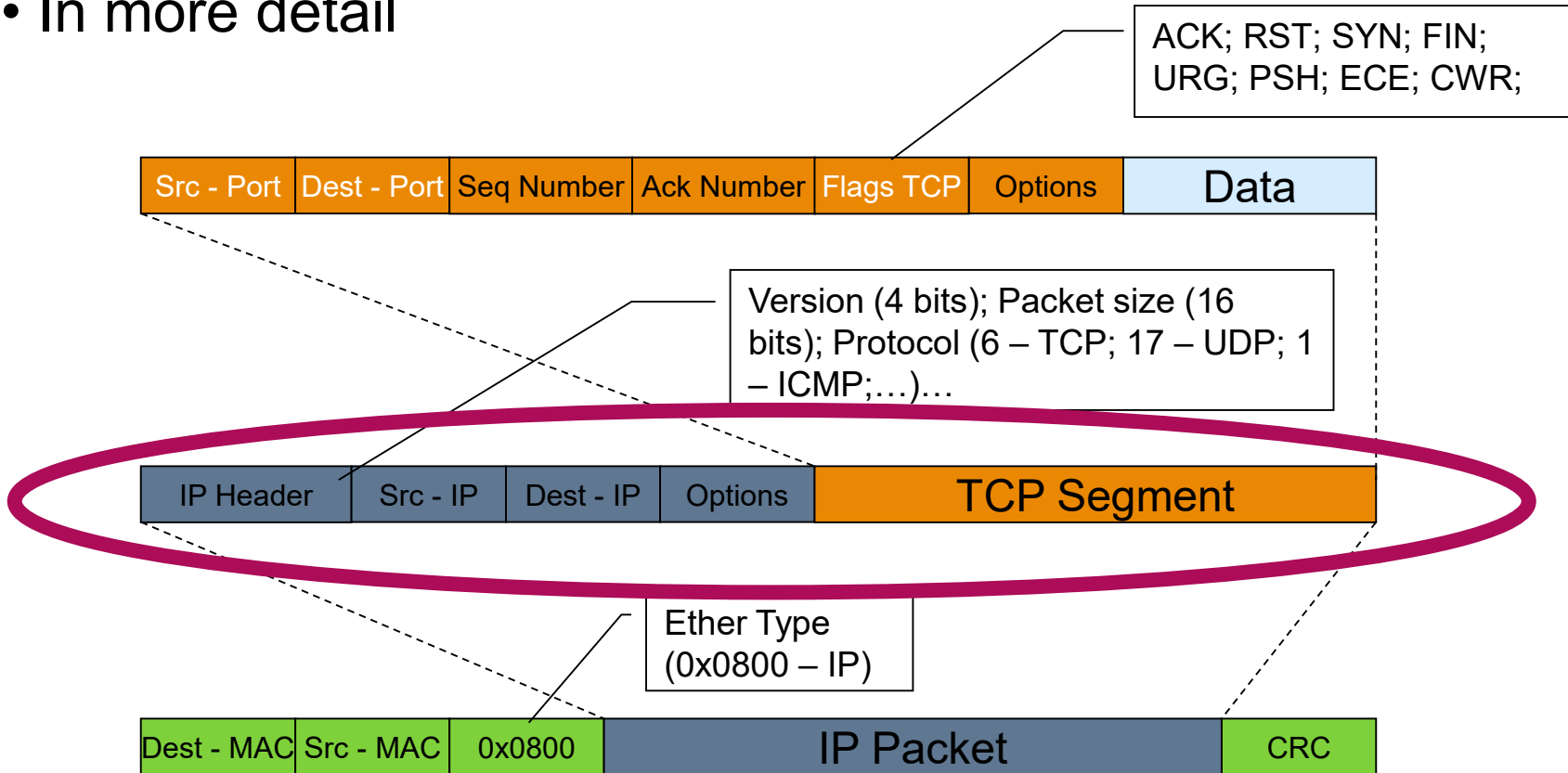
- Ethernet frame



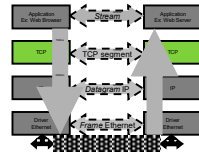
TCP/IP –Internet Communication model



- In more detail

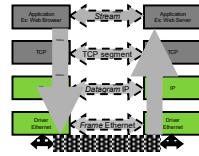


TCP/IP –Internet Communication model



- Transport Level - TCP assures a transport service:
 - connection-oriented (CO)
 - byte-stream
 - **no-loss**
 - **no-duplicates**
 - ordered

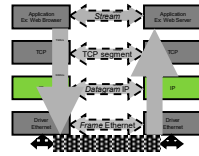
TCP/IP –Internet Communication model



■ Addresses

- Physical addresses – Media Access Controller Addresses (MAC)
 - 48 bits $\Rightarrow 2^{48}$ possible addresses ☺
- Logical addresses, or IP addresses
 - 32 bits (IPv4) $\Rightarrow 2^{32}$ possible addresses ☹
expressed in 4 groups of 8 bits each, e.g., 192.168.0.1
 - There are some reserved addresses
 - Two parts: network address and host address
 - Class A – 8 bits for network | 24 bits for hosts
maximum: 256 networks with ≈ 16 millions hosts each
 - Class B – 16 bits for network | 16 bits for hosts
maximum : ≈ 65000 networks with ≈ 65000 millions hosts each
 - Class C – 24 bits for network | 8 bits for hosts
maximum : 16 millions networks with 256 hosts each
 - Each network can still be subdivided using a **mask** (32 bits value, with 1s marking network address bits)

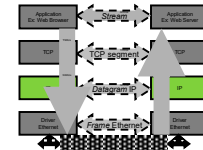
TCP/IP –Internet Communication model



- IPv4 reserved addresses

Space	Reason
0.0.0.0/8	Current Network (self-reference) addresses
10.0.0.0/8	Reserved for private networks
127.0.0.0/8	Loopback (self-address) addresses
169.254.0.0/16	Autoconfiguration (address unavailable) addresses
172.16.0.0/12	Reserved for private networks
192.0.2.0/24	Reserved for Documentation (example.com or example.net)
192.88.99.0/24	6to4 Relay Anycast Prefix (border between IPv6 and IPv4)
192.168.0.0/16	Reserved for private networks
198.18.0.0/18	Reserved for Router Input Ports
198.19.0.0/18	Reserved for Router Output Ports
224.0.0.0/4	Multicast Addresses
240.0.0.0/4	Future Use address
255.255.255.255	Limited Broadcast Address

TCP/IP –Internet Communication model



■ IPv6 addresses

- 128 bits addresses; specification [RFC 2460](#); some goals are to abolish:
 - Classes; NAT (Network Address Translation); and DHCP

... and to expand:

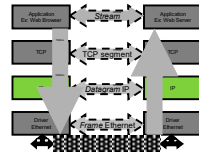
- The Net; mobility; IoT (Internet of Things)
- Format: **net_prefix** + **host_suffix** – undefined boundary



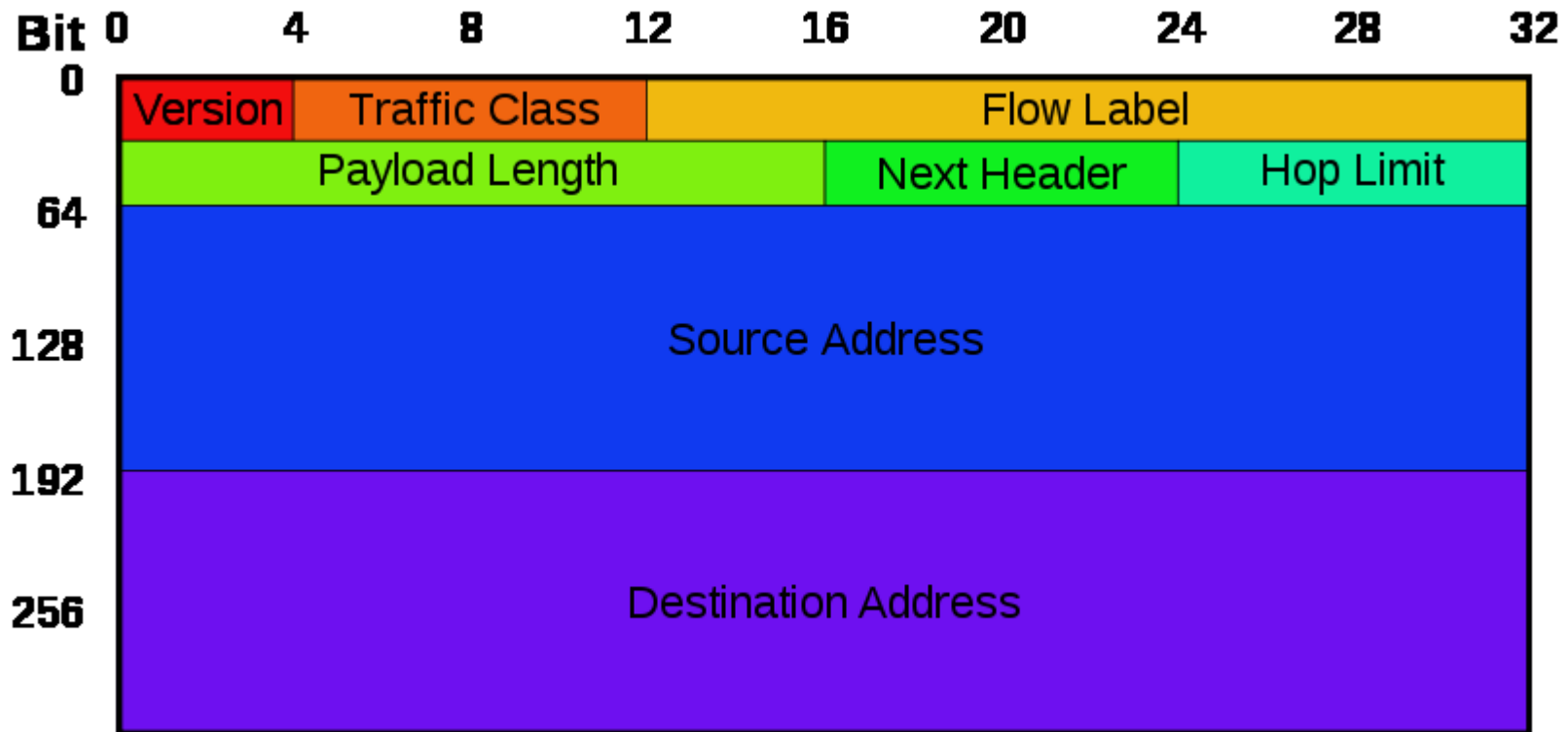
Eventually obtained from the MAC address, in EUI 64 format

- Example - > 2001:0db8:85a3:0000:**0000:0000:0000:7344**
or 2001:0db8:85a3::7344
- Address space: about $3,4 \times 10^{38}$ against about 4×10^9 of IPv4!
- Auto configuration, not requiring DHCP
- Hierarchical addressing simplifies routers implementation
- Native IPSEC support (**but not mandatory!**)
- Multicasting (ff02::1) – link-local *all nodes*
- Running concurrently with the IPv4 protocol (dual stack)
 - Reserved address for IPv4 tunnelling -> ::FFFF:<address IPv4>
 - Or encapsulating IPv6 in IPv4, using port 41!

TCP/IP –Internet Communication model

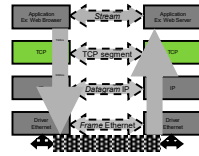


- Header format of an IPv6 datagram



<http://en.wikipedia.org/wiki/IPv6>

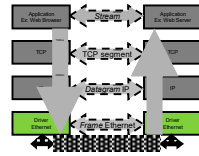
TCP/IP –Internet Communication model



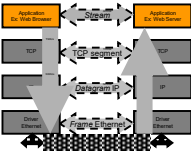
■ Addresses

- At TCP level it is much more simple. TCP and UDP headers include two 16 bits field “ports” (65.535 possible values) for origin and destination
 - Each port identifies a service
 - Historically port number from 0 to 1024 had a special meaning (for Unix processes were only accessible to the root)
 - But nowadays any application can be bind to any port ☹
 - Some well-known ports:
 - 21 (tcp) FTP
 - 23 (tcp) TELNET
 - 25 (tcp) SMTP
 - 53 (udp) DNS
 - 53 (tcp) DNS

TCP/IP –Internet Communication model



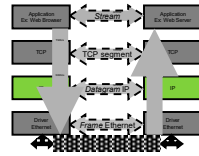
- ARP – Address Resolution Protocol
 - At link-level is it required to know the MAC address associated to a given IP!
 - An ARP packet is send (broadcast address) asking: “Who has the following IP?”... And the target IP owner node should answer with its MAC 😊
 - And if the IP address do not belong to the same network? 😞
 - All nodes have a ARP cache to maintain MAC/IP associations for all local network (minimize broadcast traffic)
 - But... facilitates *Arp Poisoning* and *ARP Flood* attacks (cache size is limited) 😞



TCP/IP –Internet Communication model

- DNS/DNSSEC – Domain Name System/Security Extensions
 - Internet machines have names! (e.g., www.uminho.pt)
 - At network-level it is necessary to resolve names for a proper IP address
 - DNS is a (**huge**) **distributed** data base that implements the above requirement
 - The network-level needs to know the IP address of its DNS server
 - Vulnerabilities and documented attacks ☹
 - <http://www.dnssec.net/dns-threats>

TCP/IP –Internet Communication model



- *Routing*: how to reach destination?
 - The most important function of the network-level
 - Each node has a **routing table**, at least with the IP address of a special node (*router* or *gateway*) linking it to the Internet
 - It is common to give routers the last host address (254 in class C networks)
 - Routers maintain routing tables to other networks and routers
 - Routers exchange routing information using two special protocols:
 - *Interior Gateway Protocols* (IGPs)
 - RIP (*Route Information Protocol*) and OSPF (*Open Shortest Path First*) are the most common ones
 - *External Gateway Protocols* (EGPs)
 - BGP (*Border Gateway Protocol*) is the most common
 - (S. Goldberg, “Why is it taking so long to secure internet routing?,” Commun. ACM, vol. 57, no. 10, pp. 56–63, 2014)



TCP/IP –Internet Communication model

- Traffic captured during execution of a browser command like <http://www.sapo.pt>

[illegible]

```
[Response In: 228]
Transaction ID: 0x613e
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
  www.sapo.pt: type A, class IN
    Name: www.sapo.pt
    [Name Length: 11]
    [Label Count: 3]
```

```
.....? I.....E.
.9?.....m.
.....5.% .[a>...
.....w ww.sapo.
pt.....
```

*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Port	Destination	Port	Protocol	Host	Info
227	2016-11-11 01:10:10.775166	192.168.1.109	51612	8.8.8.8	53	DNS		Standard query 0x613e A www.sapo.pt
228	2016-11-11 01:10:10.821428	8.8.8.8	53	192.168.1.109	51612	DNS		Standard query response 0x613e A w...
229	2016-11-11 01:10:10.826796	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [SYN] Seq=0 Win=65535 Len=...
230	2016-11-11 01:10:10.826811	192.168.1.109	49472	213.13.146.142	80	TCP		49472→80 [SYN] Seq=0 Win=65535 Len=...
231	2016-11-11 01:10:10.844832	213.13.146.142	80	192.168.1.109	49471	TCP		80→49471 [SYN, ACK] Seq=0 Ack=1 Win=...
232	2016-11-11 01:10:10.845003	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [ACK] Seq=1 Ack=1 Win=6553...
233	2016-11-11 01:10:10.845374	192.168.1.109	49471	213.13.146.142	80	HTTP	www.sapo...	GET / HTTP/1.1
234	2016-11-11 01:10:10.845771	213.13.146.142	80	192.168.1.109	49472	TCP		80→49472 [SYN, ACK] Seq=0 Ack=1 Win=...
235	2016-11-11 01:10:10.845888	192.168.1.109	49472	213.13.146.142	80	TCP		49472→80 [ACK] Seq=1 Ack=1 Win=6553...
236	2016-11-11 01:10:10.867985	213.13.146.142	80	192.168.1.109	49471	TCP		80→49471 [ACK] Seq=1 Ack=318 Win=15...
237	2016-11-11 01:10:10.869063	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]
238	2016-11-11 01:10:10.869175	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [ACK] Seq=318 Ack=1261 Win=...
239	2016-11-11 01:10:10.870046	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]
240	2016-11-11 01:10:10.870139	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [ACK] Seq=318 Ack=2521 Win=...
241	2016-11-11 01:10:10.870388	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]

[Request In: 227]
[Time: 0.046262000 seconds]
Transaction ID: 0x613e

> Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0

> Queries

> Answers

> www.sapo.pt: type A, class IN, addr 213.13.146.142
Name: www.sapo.pt
Type: A (Host Address) (1)
Class: IN (0x0001)
Time to live: 41
Data length: 4
Address: 213.13.146.142

```

0000 e0 3f 49 13 e3 93 bc 14 01 9e b6 f6 08 00 45 00 .?I.... ..E.
0010 00 49 c8 b5 00 00 2f 11 f0 c9 08 08 08 08 c0 a8 .I.../.....
0020 01 6d 00 35 c9 9c 00 35 e6 5b 61 3e 81 80 00 01 .m.5...5 .[a>...
0030 00 01 00 00 00 00 03 77 77 77 04 73 61 70 6f 02 .....w ww.sapo.
0040 70 74 00 00 01 00 01 c0 0c 00 01 00 01 00 00 00 pt.....
0050 29 00 04 d5 0d 92 8e .....).....
  
```

Text item (text), 16 bytes

Packets: 5809 · Displayed: 5809 (100.0%) Profile: Default

*Ethernet

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241	2016-11-11 01:10:10.870388	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]

▼ Transmission Control Protocol, Src Port: 49471, Dst Port: 80, Seq: 0, Len: 0

Source Port: 49471
Destination Port: 80
[Stream index: 4]
[TCP Segment Len: 0]
Sequence number: 0 (relative sequence number)
Acknowledgment number: 0
Header Length: 32 bytes

> Flags: 0x002 (SYN)
Window size value: 65535
[Calculated window size: 65535]
Checksum: 0x29d8 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0

▼ Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted

> Maximum segment size: 1260 bytes
> No-Operation (NOP)

```

0000 bc 14 01 9e b6 f6 e0 3f 49 13 e3 93 08 00 45 00 .....? I.....E.
0010 00 34 5b 5a 40 00 80 06 00 00 c0 a8 01 6d d5 0d .4[Z@... ..m..
0020 92 8e c1 3f 00 50 00 2f 3f 3c 00 00 00 00 80 02 ...?.P./ ?<.....
0030 ff ff 29 d8 00 00 02 04 04 ec 01 03 03 03 01 01 ..)... ..
0040 04 02 ..

```

TCP Options (tcp.options), 12 bytes

Packets: 5809 · Displayed: 5809 (100.0%) Profile: Default

*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Port	Destination	Port	Protocol	Host	Info
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241	2016-11-11 01:10:10.870388	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]

[Stream index: 4]
[TCP Segment Len: 0]
Sequence number: 0 (relative sequence number)
Acknowledgment number: 1 (relative ack number)
Header Length: 24 bytes
> Flags: 0x012 (SYN, ACK)
Window size value: 14600
[Calculated window size: 14600]
Checksum: 0xb043 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
▼ Options: (4 bytes), Maximum segment size
> Maximum segment size: 1460 bytes
▼ [SEQ/ACK analysis]
[This is an ACK to the segment in frame: 229]
[The RTT to ACK the segment was: 0.018056000 seconds]
[RTT: 0.018207000 seconds]

```

0000 e0 3f 49 13 e3 93 bc 14 01 9e b6 f6 08 00 45 00 .?I.....E.
0010 00 2c 00 00 40 00 38 06 19 1b d5 0d 92 8e c0 a8 ...@.8. ....
0020 01 6d 00 50 c1 3f 93 74 f0 a8 00 2f 3f 3d 60 12 .m.P.?t .../?=.
0030 39 08 b0 43 00 00 02 04 05 b4 00 00          9..C.....

```

Frame (frame), 60 bytes

Packets: 5809 · Displayed: 5809 (100.0%)

Profile: Default

Ethernet

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

Apply a display filter ... <Ctrl-/>

Expression...

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241	2016-11-11 01:10:10.870388	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]

Source Port: 49471

Destination Port: 80

[Stream index: 4]

[TCP Segment Len: 0]

Sequence number: 1 (relative sequence number)

Acknowledgment number: 1 (relative ack number)

Header Length: 20 bytes

> Flags: 0x010 (ACK)

Window size value: 65535

[Calculated window size: 65535]

[Window size scaling factor: -2 (no window scaling used)]

Checksum: 0x29cc [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

▼ [SEQ/ACK analysis]

[This is an ACK to the segment in frame: 231]

[The RTT to ACK the segment was: 0.000171000 seconds]

0000 bc 14 01 9e b6 f6 e0 3f 49 13 e3 93 08 00 45 00? I....E.

0010 00 28 5b 5c 40 00 80 06 00 00 c0 a8 01 6d d5 0d .([@... ..m..

0020 92 8e c1 3f 00 50 00 2f 3f 3d 93 74 f0 a9 50 10 ..?.P./ ?=t..P.

0030 ff ff 29 cc 00 00 ..)...

Transmission Control Protocol (tcp), 20 bytes

Packets: 5809 · Displayed: 5809 (100.0%)

Profile: Default

DSI/UM ©

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Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Expression...

No.	Time	Source	Port	Destination	Port	Protocol	Host	Info
230	2016-11-11 01:10:10.826811	192.168.1.109	49472	213.13.146.142	80	TCP		49472→80 [SYN] Seq=0 Win=65535 Len=...
231	2016-11-11 01:10:10.844832	213.13.146.142	80	192.168.1.109	49471	TCP		80→49471 [SYN, ACK] Seq=0 Ack=1 Win=...
232	2016-11-11 01:10:10.845003	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [ACK] Seq=1 Ack=1 Win=6553...
233	2016-11-11 01:10:10.845374	192.168.1.109	49471	213.13.146.142	80	HTTP	www.sapo...	GET / HTTP/1.1
234	2016-11-11 01:10:10.845771	213.13.146.142	80	192.168.1.109	49472	TCP		80→49472 [SYN, ACK] Seq=0 Ack=1 Win=...
235	2016-11-11 01:10:10.845888	192.168.1.109	49472	213.13.146.142	80	TCP		49472→80 [ACK] Seq=1 Ack=1 Win=6553...
236	2016-11-11 01:10:10.867985	213.13.146.142	80	192.168.1.109	49471	TCP		80→49471 [ACK] Seq=1 Ack=318 Win=15...

[SEQ/ACK analysis]

[iRTT: 0.018207000 seconds]

[Bytes in flight: 317]

[Bytes sent since last PSH flag: 317]

Hypertext Transfer Protocol

GET / HTTP/1.1\r\n

[Expert Info (Chat/Sequence): GET / HTTP/1.1\r\n]

[GET / HTTP/1.1\r\n]

[Severity level: Chat]

[Group: Sequence]

Request Method: GET

Request URI: /

Request Version: HTTP/1.1

Accept: text/html, application/xhtml+xml, image/jxr, */*\r\n

Accept-Language: en-GB\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.14393\r\n

Accept-Encoding: gzip, deflate\r\n

Host: www.sapo.pt\r\n

Connection: Keep-Alive\r\n

\r\n

[Full request URI: http://www.sapo.pt/]

[HTTP request 1/8]

[Response in frame: 290]

[Next request in frame: 297]

0090	20 65 6e 2d 47 42 0d 0a	55 73 65 72 2d 41 67 65	en-GB.. User-Age
00a0	6e 74 3a 20 4d 6f 7a 69	6c 6c 61 2f 35 2e 30 20	nt: Mozi lla/5.0
00b0	28 57 69 6e 64 6f 77 73	20 4e 54 20 31 30 2e 30	(Windows NT 10.0
00c0	3b 20 57 69 6e 36 34 3b	20 78 36 34 29 20 41 70	; Win64; x64) Ap
00d0	70 6c 65 57 65 62 4b 69	74 2f 35 33 37 2e 33 36	pleWebKi t/537.36
00e0	20 28 4b 48 54 4d 4c 2c	20 6c 69 6b 65 20 47 65	(KHTML, like Ge

HTTP User-Agent header (http.user_agent), 142 bytes

Packets: 5809 · Displayed: 5809 (100.0%)

Profile: Default

DSI/UM ©

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*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Port	Destination	Port	Protocol	Host	Info
232	2016-11-11 01:10:10.845003	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [ACK] Seq=1 Ack=1 Win=6553...
233	2016-11-11 01:10:10.845374	192.168.1.109	49471	213.13.146.142	80	HTTP	www.sapo...	GET / HTTP/1.1
234	2016-11-11 01:10:10.845771	213.13.146.142	80	192.168.1.109	49472	TCP		80→49472 [SYN, ACK] Seq=0 Ack=1 Win...
235	2016-11-11 01:10:10.845888	192.168.1.109	49472	213.13.146.142	80	TCP		49472→80 [ACK] Seq=1 Ack=1 Win=6553...
236	2016-11-11 01:10:10.867985	213.13.146.142	80	192.168.1.109	49471	TCP		80→49471 [ACK] Seq=1 Ack=318 Win=15...
237	2016-11-11 01:10:10.869063	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]
238	2016-11-11 01:10:10.869175	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [ACK] Seq=318 Ack=1261 Win...

> Frame 236: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

> Ethernet II, Src: HitronTe_9e:b6:f6 (bc:14:01:9e:b6:f6), Dst: AsustekC_13:e3:93 (e0:3f:49:13:e3:93)

> Internet Protocol Version 4, Src: 213.13.146.142, Dst: 192.168.1.109

▼ Transmission Control Protocol, Src Port: 80, Dst Port: 49471, Seq: 1, Ack: 318, Len: 0

Source Port: 80

Destination Port: 49471

[Stream index: 4]

[TCP Segment Len: 0]

Sequence number: 1 (relative sequence number)

Acknowledgment number: 318 (relative ack number)

Header Length: 20 bytes

> Flags: 0x010 (ACK)

Window size value: 15544

[Calculated window size: 15544]

[Window size scaling factor: -2 (no window scaling used)]

Checksum: 0xc313 [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

▼ [SEQ/ACK analysis]

[\[This is an ACK to the segment in frame: 233\]](#)

[The RTT to ACK the segment was: 0.022611000 seconds]

[iRTT: 0.018207000 seconds]

```

0000  e0 3f 49 13 e3 93 bc 14 01 9e b6 f6 08 00 45 00  .?I.....E.
0010  00 28 99 1f 40 00 38 06 7f ff d5 0d 92 8e c0 a8  .(..@.8. ....
0020  01 6d 00 50 c1 3f 93 74 f0 a9 00 2f 40 7a 50 10  .m.P?.t .../zP.
0030  3c b8 c3 13 00 00 00 00 00 00 00 00 00 00 00 00  <.....

```

wireshark_B8350999-BD9B-4020-BE61-9665612E39BA_20161111011007_a01144

Packets: 5809 · Displayed: 5809 (100.0%) Profile: Default

*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Port	Destination	Port	Protocol	Host	Info
231	2016-11-11 01:10:10.844832	213.13.146.142	80	192.168.1.109	49471	TCP		80→49471 [SYN, ACK] Seq=0 Ack=1 Win...
232	2016-11-11 01:10:10.845003	192.168.1.109	49471	213.13.146.142	80	TCP		49471→80 [ACK] Seq=1 Ack=1 Win=6553...
233	2016-11-11 01:10:10.845374	192.168.1.109	49471	213.13.146.142	80	HTTP	www.sapo...	GET / HTTP/1.1
234	2016-11-11 01:10:10.845771	213.13.146.142	80	192.168.1.109	49472	TCP		80→49472 [SYN, ACK] Seq=0 Ack=1 Win...
235	2016-11-11 01:10:10.845888	192.168.1.109	49472	213.13.146.142	80	TCP		49472→80 [ACK] Seq=1 Ack=1 Win=6553...
236	2016-11-11 01:10:10.867985	213.13.146.142	80	192.168.1.109	49471	TCP		80→49471 [ACK] Seq=1 Ack=318 Win=15...
237	2016-11-11 01:10:10.869063	213.13.146.142	80	192.168.1.109	49471	TCP		[TCP segment of a reassembled PDU]

[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0x769b [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
[SEQ/ACK analysis]
[iRTT: 0.018207000 seconds]
[Bytes in flight: 1260]
[Bytes sent since last PSH flag: 1260]
TCP segment data (1260 bytes)

```

0000 e0 3f 49 13 e3 93 bc 14 01 9e b6 f6 08 00 45 00 .?I.....E.
0010 05 14 99 20 40 00 38 06 7b 12 d5 0d 92 8e c0 a8 ... @.8. {.....
0020 01 6d 00 50 c1 3f 93 74 f0 a9 00 2f 40 7a 50 10 .m.P?.t .../@zP.
0030 3c b8 76 9b 00 00 48 54 54 50 2f 31 2e 31 20 32 <.v...HT TP/1.1 2
0040 30 30 20 4f 4b 0d 0a 53 65 72 76 65 72 3a 20 53 00 OK..S erver: S
0050 41 50 4f 74 74 70 64 2f 32 2e 30 0d 0a 56 61 72 APottpd/ 2.0..Var
0060 79 3a 20 41 63 63 65 70 74 2d 45 6e 63 6f 64 69 y: Accep t-Encodi
0070 6e 67 0d 0a 58 2d 53 65 72 76 65 72 3a 20 4b 61 ng..X-Se rver: Ka
0080 72 61 20 4b 75 6d 0d 0a 58 2d 46 72 61 6d 65 2d ra Kum.. X-Frame-
0090 4f 70 74 69 6f 6e 73 3a 20 44 45 4e 59 0d 0a 58 Options: DENY..X
00a0 2d 55 41 2d 43 6f 6d 70 61 74 69 62 6c 65 3a 20 -UA-Comp atible:
00b0 49 45 3d 65 64 67 65 0d 0a 43 61 63 68 65 2d 43 IE=edge. .Cache-C
00c0 6f 6e 74 72 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 ontrol: no-cache
00d0 2c 20 6e 6f 2d 73 74 6f 72 65 0d 0a 50 72 61 67 , no-sto re..Prag
00e0 6d 61 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 53 65 ma: no-c ache..Se
00f0 74 2d 43 6f 6f 6b 69 65 3a 20 75 75 3d 35 65 36 t-Cookie : uu=5e6
0100 36 39 37 39 37 65 33 64 31 34 64 64 32 39 30 39 69797e3d 14dd2909
0110 62 61 35 39 36 31 31 39 61 66 38 61 64 3b 20 4d ba596119 af8ad; M
0120 61 78 2d 41 67 65 3d 33 36 30 30 3b 20 45 78 70 ax-Age=3 600; Exp
0130 69 72 65 73 3d 46 72 69 2c 20 31 31 20 4e 6f 76 ires=Fri , 11 Nov
0140 20 32 30 31 36 20 30 32 3a 31 30 3a 31 30 20 47 2016 02 :10:10 G
0150 4d 54 3b 20 50 61 74 68 3d 2f 0d 0a 43 6f 6e 74 MT; Path =/..Cont
0160 65 6e 74 2d 54 79 70 65 3a 20 74 65 78 74 2f 68 ent-Type : text/h

```

A data segment used in reassembly of a lower-level protocol (tcp.segment_data), 1260 bytes

Packets: 5809 · Displayed: 5809 (100.0%) Profile: Default

*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 2

No.	Time	Source	Port	Destination	Port	Protocol	Host	Info
206	2016-11-11 01:10:10.442396	204.79.197.200	443	192.168.1.109	49458	TCP		[TCP segment of a reassembled PDU]
207	2016-11-11 01:10:10.442396	204.79.197.200	443	192.168.1.109	49458	TCP		[TCP segment of a reassembled PDU]
208	2016-11-11 01:10:10.442397	204.79.197.200	443	192.168.1.109	49458	TLSv1.2		Application Data
209	2016-11-11 01:10:10.442471	192.168.1.109	49458	204.79.197.200	443	TCP		49458→443 [ACK] Seq=6948 Ack=68014 ...
214	2016-11-11 01:10:10.548680	192.168.1.109	49458	204.79.197.200	443	TLSv1.2		Application Data
215	2016-11-11 01:10:10.555063	192.168.1.109	49458	204.79.197.200	443	TCP		49458→443 [FIN, ACK] Seq=7929 Ack=6...
217	2016-11-11 01:10:10.570204	204.79.197.200	443	192.168.1.109	49458	TCP		443→49458 [ACK] Seq=68014 Ack=7929 ...

Ethernet II, Src: AsustekC_13:e3:93 (e0:3f:49:13:e3:93), Dst: HitronTe_9e:b6:f6 (bc:14:01:9e:b6:f6)

Internet Protocol Version 4, Src: 192.168.1.109, Dst: 204.79.197.200

Transmission Control Protocol, Src Port: 49458, Dst Port: 443, Seq: 7929, Ack: 68014, Len: 0

Source Port: 49458
Destination Port: 443
[Stream index: 2]
[TCP Segment Len: 0]
Sequence number: 7929 (relative sequence number)
Acknowledgment number: 68014 (relative ack number)
Header Length: 20 bytes

Flags: 0x011 (FIN, ACK)
Window size value: 32768
[Calculated window size: 32768]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x5448 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0

```

0000  bc 14 01 9e b6 f6 e0 3f 49 13 e3 93 08 00 45 00  .....? I.....E.
0010  00 28 4e 7c 40 00 80 06 00 00 c0 a8 01 6d cc 4f  .(N|@... ..m.O
0020  c5 c8 c1 32 01 bb 0b fd 84 2f ca 47 f1 4d 50 11  ..2....../.G.MP.
0030  80 00 54 48 00 00  ..TH..

```

Transmission Control Protocol (tcp), 20 bytes

Packets: 5809 · Displayed: 103 (1.8%) · Dropped: 0 (0.0%) Profile: Default

*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 2

No.	Time	Source	Port	Destination	Port	Protocol	Host	Info
209	2016-11-11 01:10:10.442471	192.168.1.109	49458	204.79.197.200	443	TCP		49458→443 [ACK] Seq=6948 Ack=68014 ...
214	2016-11-11 01:10:10.548680	192.168.1.109	49458	204.79.197.200	443	TLSv1.2		Application Data
215	2016-11-11 01:10:10.555063	192.168.1.109	49458	204.79.197.200	443	TCP		49458→443 [FIN, ACK] Seq=7929 Ack=6...
217	2016-11-11 01:10:10.570204	204.79.197.200	443	192.168.1.109	49458	TCP		443→49458 [ACK] Seq=68014 Ack=7929 ...
218	2016-11-11 01:10:10.574299	204.79.197.200	443	192.168.1.109	49458	TCP		443→49458 [ACK] Seq=68014 Ack=7930 ...
219	2016-11-11 01:10:10.575295	204.79.197.200	443	192.168.1.109	49458	TLSv1.2		Application Data
220	2016-11-11 01:10:10.575338	192.168.1.109	49458	204.79.197.200	443	TCP		49458→443 [RST, ACK] Seq=7930 Ack=6...

> Ethernet II, Src: AsustekC_13:e3:93 (e0:3f:49:13:e3:93), Dst: HitronTe_9e:b6:f6 (bc:14:01:9e:b6:f6)

> Internet Protocol Version 4, Src: 192.168.1.109, Dst: 204.79.197.200

> Transmission Control Protocol, Src Port: 49458, Dst Port: 443, Seq: 7930, Ack: 68547, Len: 0

Source Port: 49458

Destination Port: 443

[Stream index: 2]

[TCP Segment Len: 0]

Sequence number: 7930 (relative sequence number)

Acknowledgment number: 68547 (relative ack number)

Header Length: 20 bytes

> Flags: 0x014 (RST, ACK)

Window size value: 0

[Calculated window size: 0]

[Window size scaling factor: -1 (unknown)]

Checksum: 0x5448 [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

```

0000 bc 14 01 9e b6 f6 e0 3f 49 13 e3 93 08 00 45 00 .....? I.....E.
0010 00 28 4e 7e 40 00 80 06 00 00 c0 a8 01 6d cc 4f .(N~@... ..m.O
0020 c5 c8 c1 32 01 bb 0b fd 84 30 ca 47 f3 62 50 14 ..2.... .0.G.bP.
0030 00 00 54 48 00 00 ..TH..

```

Transmission Control Protocol (tcp), 20 bytes

Packets: 5809 · Displayed: 103 (1.8%) · Dropped: 0 (0.0%) Profile: Default

TCP/IP –Internet Communication model

- Signs of anomalies (TCP)
 - Attempt to establish a connection without fulfil the three handshaking phases (*SYN Flooding*)
 - A large number of ICMP packets (*PING Flooding*)
 - Access to unexpected ports
 - After establishing a connection transferring few data is very suspect
 - Verify who started connections (*IP spoofing!*)
 - High level utilization of ARP (*ARP spoofing / Session hijack*)
 - ACK packets out of sequence (computers usually answer with a RESET, indicating the have the port oppended 😊)
 - SYN packets with data 😞
 - The standard allows it, but it is highly suspect!!!
 - Sometimes used by legitimate services like DNS (performance reasons)



TCP/IP –Internet Communication model

- Tools for traffic capture and analysis
 - Capture and decoding
 - **tcpdump** (windump) – based on **libpcap** that implements BPF model (BSD Packet Filter)
 - **dumppcap** – variant of the above
 - Idem + protocol analysis
 - **wireshark** (GUI) / **tshark** (TTY)
 - **analyzer**
 - These tools make part of security *tool kits* (compilations), like NST (Network Security Tools) or **Kali** (ex-Back Track)
 - Many other tools:
 - <http://www.slac.stanford.edu/xorg/nmtf/nmtf-tools.html>



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